KU Leuven

Faculty of Engineering Technology

Your Thesis Title Here

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Abstract

Acknowledgements

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Nomenclature

Symbols

$\boldsymbol{q}$	Joint position vector	[rad]
K_s	Spring stiffness	[Nm/rad]

Abbreviations

CAN	Controller Area Network
SET	Series Elastic Transmission

Chapter 1

Introduction

1.1 Background and Motivation

The musculoskeletal system contains muscles, bones, joints and connective tissues like tendons and ligaments surrounding these structures. Musculoskeletal disorders (MSDs) can therefore be defined as injuries of the muscles, nerves, tendons, joints, cartilage, and spinal discs. When these injuries are sustained or worsened by exposure to risk factors in the workspace, they are also known as work-related musculoskeletal disorders (WMSDs). These disorders can go from minor aches to severe pain and chronic illness needing medical treatment, resulting in workers not being able to do physical jobs.

A degenerating physical state of the body can also result in mental health issues like depression or frustration causing low job satisfaction, which will indirectly cause workers to perform worse even if they physically can keep working.

WMSDs are developed over time due to non-ergonomic working conditions and can be categorized into two main groups:

- Back disorders related to handling loads
- Upper limb disorders, also known as repetitive strain injuries (RSIs)

MSDs can be caused by for example:

- Twisting and bending movements while handling loads
- Repetitive or forceful movements
- Awkward and static postures
- Vibration
- Prolonged sitting or standing in the same position

A large increase in MSDs has been reported worldwide. This rise is partly attributed to improved recognition and reporting, but also to the tendency for work to be more segmented. This tendency arises from an increasingly competitive global economy where fast-pased repetitive work environments become more widespread.

It is interesting to note that the increase in education in the workforce, combined with increased expectations regarding job satisfaction, has reduced workers' tolerance for WMSDs. An ergonomic work environment can prevent the development of WMSDs and directly reduce the cost to companies due to sick leave, as well as reducing the societal impact. Ergonomics can be defined as the study of work, where the goal is to fit tasks and jobs to workers'

capabilities, instead of forcing workers to fit the job. An ergonomic work environment is thus one that implements task-specific strategies to prevent reduced work ability. Reduced work ability can be categorized in absenteeism and presenteeism. Absenteeism refers to the absence from work due to injuries sustained on the job. Presenteeism describes the situation in which workers continue to attend their expected working duties despite existing injuries, often due to concerns about job security. In such cases, work is performed in a self–protective manner to reduce physical strain.

Musculoskeletal disorders affect approximately one in three people worldwide. Within the European Union, MSDs represent the most costly health-related problems, affecting 45 million workers, many of whom are unable to continue working after the age of 60 due to these diseases. This results in billions of euros in lost revenue for companies and a great burden on the public healthcare system.

In Belgium, 2.5 million people were affected by MSDs in 2018 (approximately 62% of the Belgian working population in 2015) resulting in 3 billion euros in direct medical costs and 2 billion in work absenteeism, which is 0.3% of the Belgian gross domestic product [1][2][3][4][5].

There is a clear need to address reduced work efficiency resulting from physical injuries since this improves the quality of life of an increasingly aging workforce, but also to reduce the impact on social welfare systems and increase overall economic productivity. Yet few collaborative applications exist in production facilities. Human and robots are either physically seperated, or work in a coexistence arrangement where no physical barrier seperates human and robot but no direct interaction takes place. Implementations like this are only economically viable for large batch sizes, since a whole production line needs to be tuned towards that specific robotic solution. Cage–free robotic arms that share a workspace with a human worker are introduced to increase flexibility of the production line. Interactive robotic solutions for small batches and varying product sizes has become a clear need since 70% of collaboration between human worker and robotic solutions are found in assembly applications. An industry survey reported that 86% of respondents were willing to invest in lightweight robotic solutions, indicating strong industrial interest. The primary motivations for introducing robotic arms in assembly environments are:

- Improving operational efficiency
- Improving ergonomics

Cage–free robotic solutions generally involve higher initial costs compared to traditional fenced systems due to safety certification requirements and CE conformity procedures. Companies are however often willing to accept longer payback periods for implementing and testing company specific robotic solutions which increase ergonomics. Traditional stiff robots in collaborative applications are heavily regulated to minimise the risk of injuries. These robots are lightweight, operate at low speeds and often lack mechanical bandwidth to react to unforeseen collisions in a safe and compliant way[6].

One solution for safe collaboration between human and robot are hand–guided robotic arms. Hand–guided robotic applications are defined as a direct physical interaction between human and robot, where the human operator has full control over the robot motion. In this

application, a direct input device is located at or near the end–effector to measure input forces and simultaneously define the position of the human in the shared workspace. A direct consequence of this approach is that robotic applications with sensorised end–effectors can only react in a passive compliant manner when handled at a specific point on the robotic arm. Collisions or attemted handling of the robotic arm elsewhere will result in a safety stop of the robotic arm[7].

Existing robotic solutions that allow a human worker to hand–guide a robotic arm by grapsing it wherever is most convenient are, to the best of the autors knowledge, practically non-existent in assembly and packaging industries. When hand–guiding is discussed in literature, it is typically limited to sensorised end–effectors on conventional stiff robots[8][9].

This absence in current industrial applications is further supported by surveys of end users, system integrators and researchers which states that collaborative robots are still primarily implemented as traditional stiff robots with limited or no direct human interaction [10].

Example bold paragraph title

Your text here **somekey**.

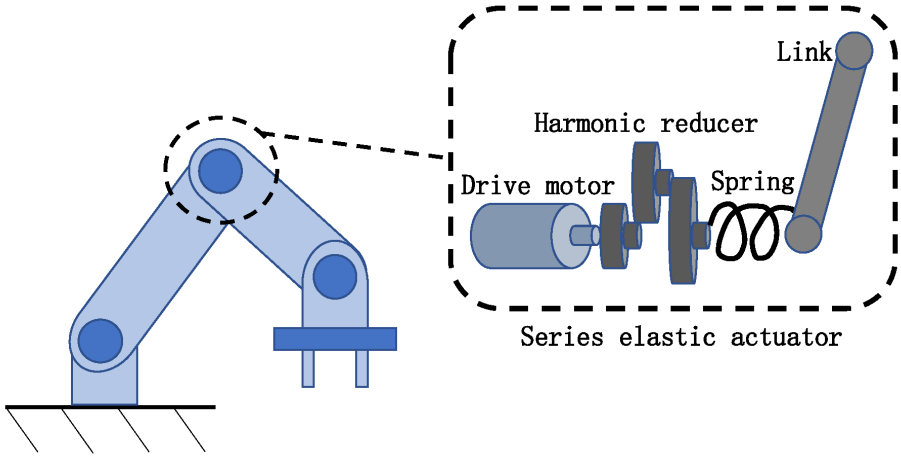

Figure 1.1: Caption text.

1.2 Problem Statement

1.3 Thesis Outline

Chapter 2

Literature Review

2.1 Theoretical Background

2.1.1 Example subsection

Example equation:

$$\tau = K_s \, \theta_s \tag{2.1}$$

2.1.2 Another subsection

2.2 State of the Art

2.3 Gap Analysis

Chapter 3

System Design and Implementation

3.1 System Overview and Requirements

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3.5.3 Weighing Mode

3.5.4 Carrying Mode

3.5.5 Put-Down Mode

3.6 Software Implementation

3.6.1 Real-Time Architecture

Example code listing:

```
while (running) {
    clock_nanosleep(...);
    // your code here
}
```

Listing 3.1: Example listing.

3.6.2 Configuration

Chapter 4

Validation and Results

4.1 Experimental Setup

4.1.1 Hardware Configuration

4.1.2 Parameter Identification

4.2 Spring and Sensor Characterisation

4.3 Controller Performance

4.4 Assisted Load Handling Evaluation

4.5 Discussion

Example table:

Table 4.1: Example results table.

Metric	Target	Achieved
Value 1	X	Y
Value 2	X	Y

Chapter 5

Conclusion and Future Work

5.1 Conclusion

5.2 Future Work

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2D Manufacturing Drawings

.1 CNC Part 1

.2 CNC Part 2

.3 CNC Part 3

Additional Experimental Results

Source Code

https://github.com/username/repository